

Ziyi Wang

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Education

Georgia Institute of Technology

Atlanta, GA

M.S. Robotics (Expected May 2028)

University of Illinois at Urbana-Champaign

Urbana, IL

B.S. Computer Engineering (May 2026, Highest Honors)

Dean's List. **Coursework:** Robot Dynamics and Control, Control Systems, Deep Learning for Computer Vision, Computer Systems Engineering. **Graduate research:** Robot Learning and Reasoning Lab, advised by Prof. Danfei Xu.

Summary

First-year Georgia Tech M.S. Robotics student conducting research in the Robot Learning and Reasoning Lab, advised by Prof. Danfei Xu, with experience in Isaac Lab vision-tactile simulation, humanoid teleoperation, RGB-D perception, and onboard perception-to-control deployment.

Skills

Programming / ML: Python, C++, C, MATLAB, PyTorch, OpenCV, scikit-learn

Robot Learning / Human Data: Isaac Lab, multi-environment RL workflows, vision-tactile simulation, teleoperation trajectory collection, Gazebo

Perception / Robot Systems: RGB-D perception, 6D pose estimation, FoundationPose, YOLO, ROS 2 / ROS Noetic, NVIDIA Jetson, RealSense

Control / Manipulation: IK/FK, MoveIt, trajectory planning, visual servoing, teleoperation, PID, MPPI

Systems / Embedded: Linux, Docker, Git, STM32, FreeRTOS, CAN, USART/UART, SPI, I2C, PWM

Research and Robotics Experience

Robot Learning Engineer Intern

Jun 2026 – Present

BrainCo

Shenzhen, China

- Building Isaac Lab multi-environment RL training workflows for dexterous-hand vision-tactile simulation, bringing up parallel training setup and tactile simulation modalities for ongoing robot-learning experiments.
- Training a tactile adaptation layer from hardware-team feedback to explore sim-to-hardware signal alignment; benchmarking parallel simulation configurations by maximum environment count and per-step simulation time.

RGB-D Perception for Wheeled Humanoid Teleoperation

Aug 2024 – May 2025

Undergraduate Researcher, RoboDesign Lab (Advisor: Prof. Joao Ramos)

UIUC

- Deployed pre-trained CenterSnap ONNX models on a Jetson Orin Nano as SATYRR's vision-based object size estimator; captured 640x480 / 30 Hz aligned RealSense RGB-D, clipped depth at 2 m, and streamed length/width/height priors to the controller PC over UDP.
- Added per-frame max-pooling, a 4-frame sliding-window buffer, and IQR temporal outlier rejection, improving size accuracy by 15% vs. raw single-frame output and reducing downstream inertia-estimation convergence time by 5% on a benchmark object.
- Operated SATYRR HMI to record bilateral-teleoperation trajectories and ran perception-in-the-loop lifting/delivery sessions for multi-stage object-parameter estimation experiments.

Vision-Based Autonomous Perching

May 2025 – Sep 2025

Undergraduate Researcher, Advanced Control Research Lab (Advisor: Prof. Naira Hovakimyan)

UIUC

- Co-authored “Grasping as Anchoring: Expanding Urban Perching with a Hybrid Gripper for Aerial Robots” (Han et al., accepted to IEEE/RSJ IROS, 5th of 9 authors); contributed the onboard 6D perception stack on NVIDIA Jetson Orin.
- Wrote a ROS 2 wrapper around FoundationPose for 20 Hz texture-CAD-based 6D pose tracking using a forward-view RealSense D435i; published `camera→object` TF consumed by trajectory generation and $SE(3)+\mathcal{L}_1$ Quad control before 10–30 Hz terminal visual servoing.
- Built an offline text-prompted zero-shot segmentation pipeline (GroundingDINO + SAM2 via HuggingFace) for FoundationPose initialization; validated end-to-end on 88 perching trials across 8 pole-head geometries, achieving 88.6% success vs. 10% with visual servoing disabled.

Embedded Team Lead, Illini RoboMaster

Jan 2023 – May 2026

Student Robotics Organization

UIUC

- Led embedded C/C++ firmware and validation for 2 competition robots used by 15 team members, supporting 20 motors/CAN devices across chassis, gimbal, and actuator subsystems.
- Applied IK/FK and MoveIt motion planning for 6-DOF robotic arm teleoperation, bridging operator commands, USART/PPM/DBUS joint-angle transmission, and planned manipulator motion.

Awards

RoboMaster North America 2023 1v1 2nd Place (Top 10%); 2025–2026 Ellery B. Paine Outstanding Junior Award; Yunni and Maxine Pao Memorial Scholarship; John and Sheila Woythal Scholarship.